Asymmetric Iron-Ore Shock Transmission in Australian Mining Equities:

A Nonlinear ARDL and Regime-Switching Analysis

Kunal Arora

Department of Commerce, Delhi School of Economics, University of Delhi
workkunal1@commerce.du.ac.in

June 2026

SSRN Working Paper

This paper has not been peer reviewed.
Comments and suggestions are welcome.

Abstract

We examine whether iron-ore price shocks transmit asymmetrically to the equity returns of Australia's three largest listed iron-ore producers (BHP, Rio Tinto, Fortescue) over January 2011 to May 2026 (N = 3,013 trading days). Using a nonlinear ARDL (NARDL) framework, we find that positive iron-ore price movements transmit substantially more strongly to mining-equity prices than negative movements: the long-run multiplier on cumulative price increases ($L^+ = 0.720$) exceeds that on decreases ($L^- = 0.523$), a gap of 0.197 that is statistically significant (Wald $F = 13.90$, $p < 0.001$). The asymmetry is robust across alternative market-control specifications and lag structures, and is sharply concentrated by business model: the pure-play producer (Fortescue) exhibits long-run multipliers three to four times those of the diversified majors. Evidence on a long-run cointegrating relationship is suggestive rather than conclusive—the bounds F-statistic is inconclusive at the 5% level, though a significantly negative error-correction term and the bounds t-test point toward such a relationship. A two-state Markov-switching model identifies persistent calm and stress regimes; embedding the regime classification in the NARDL via filtered probabilities reveals a two-part structure: the long-run asymmetry is regime-invariant (the calm–stress difference in the asymmetry gap is statistically indistinguishable from zero, and survives a two-stage block bootstrap that propagates first-stage regime-estimation uncertainty), whereas short-run transmission intensity roughly triples in the stress regime. These findings indicate that the asymmetric structure of commodity–equity transmission is a stable feature of the relationship, while its immediate magnitude is state-dependent. The results imply that hedges built on symmetric, constant exposure are systematically miscalibrated, that the three producers are distinct rather than interchangeable instruments for iron-ore exposure, and that hedge levels can rest on stable long-run parameters while rebalancing intensity should rise in stressed markets.

Keywords: nonlinear ARDL; asymmetric transmission; iron ore; mining equities; Markov switching; commodity–equity linkages

JEL classification: G12; G15; Q02; C22

1. Introduction

Australia is the world's largest exporter of iron ore, and a small number of listed producers—BHP, Rio Tinto, and Fortescue Metals Group—account for the bulk of national production and a material share of the country's equity-market capitalisation. For these firms, the price of iron ore is the single most important fundamental driver of cash flows. Yet the way iron-ore price shocks transmit into their equity valuations need not be symmetric. Investors may price an iron-ore price rise differently from an equal-sized fall, because of operating leverage, the option-like character of equity in firms with fixed extraction costs, asymmetric expectations about the persistence of price moves, or differences in how good and bad commodity news is incorporated into prices.

This paper asks three questions. First, does iron-ore price transmission to Australian mining equities exhibit long-run asymmetry between price increases and decreases? Second, does the strength of that transmission scale with a firm's exposure to iron ore—so that a pure-play producer responds more than diversified miners? Third, is the asymmetry a stable structural feature of the relationship, or does it change between calm and turbulent market states?

These questions sit in a literature that has, to date, looked elsewhere. The nonlinear ARDL approach to asymmetric commodity–equity transmission has been applied extensively to oil, gold and exchange rates—for example to oil-price effects on Asian equity indices (Truong et al., 2024a, 2024b) and on renewable-energy stocks (Mishra and Acharya, 2025)—but iron ore has received comparatively little attention through this lens, despite being Australia's dominant export commodity and the primary fundamental for its largest listed firms. We are, moreover, not aware of work that pairs the asymmetry question with a regime-switching layer to ask whether the asymmetry is a structural feature of the relationship or a creature of crisis periods. This paper addresses that combination: it brings the NARDL framework to iron-ore transmission specifically, examines how the asymmetry varies with a firm's commodity concentration, and embeds a Markov-switching regime classification to separate the structural from the state-dependent. We frame this as positioning rather than a claim of priority—the value is in the specific combination and the resulting two-part characterisation, not in being first to any one element.

The engine for all three questions is a nonlinear ARDL (NARDL) model in the tradition of Shin, Yu and Greenwood-Nimmo (2014). The NARDL decomposes cumulative iron-ore price changes into separate partial sums of increases and decreases, allowing positive and negative movements to carry distinct long-run multipliers while retaining a single cointegrating error-correction structure. The framework has been applied extensively to commodity–equity relationships in recent work: Truong et al. (2024a, 2024b) document asymmetric oil-price effects on Vietnamese equity indices, and Mishra and Acharya (2025) find that oil-price asymmetry in Indian renewable-energy stock returns is concentrated in firms with the most direct commodity exposure—a firm-level pattern we revisit for iron ore below. To address the third question, we combine the NARDL with a two-state Markov-switching model of the return-generating process, and embed the resulting regime classification into the NARDL through a dummy-interaction specification estimated on the full continuous sample.

Our central finding is a robust long-run asymmetry: a sustained iron-ore price increase raises the mining-equity portfolio's long-run equilibrium value by more than an equally sized decrease lowers it, with long-run multipliers of $L^+ = 0.720$ versus $L^- = 0.523$. The 0.197 gap is highly significant (Wald $F = 13.90$, $p < 0.001$) and stable across market controls and lag choices. The effect is concentrated by business model: Fortescue, a pure-play iron-ore producer, shows long-run multipliers three to four times those of the diversified majors BHP and Rio Tinto.

We make two further contributions to how this relationship is characterised. First, we are careful about the strength of the cointegration evidence: under the correct critical values for our model the bounds F-test is inconclusive at the 5% level, and we therefore rest the long-run interpretation on the combination of a significantly negative error-correction coefficient and a bounds t-test that clears its lower threshold—a more honest reading than a single marginal F-statistic would support. Second, the regime analysis yields a clean two-part result: the long-run asymmetry is regime-invariant—statistically indistinguishable between calm and stress states, and robust to a two-stage block bootstrap that propagates the uncertainty in the estimated regime classification—whereas the short-run intensity of transmission roughly triples in the stress regime. The asymmetric structure is thus a stable property of the relationship, while its immediate magnitude is state-dependent.

The remainder of the paper is organised as follows. Section 2 describes the data. Section 3 sets out the NARDL methodology and the regime-switching extension. Section 4 reports the main results and robustness checks. Section 5 presents the regime-conditional analysis. Section 6 draws out the economic significance and practical implications of the findings. Section 7 discusses diagnostics and the steps taken to address them. Section 8 sets out limitations and directions for future research. Section 9 concludes.

2. Data

The sample runs from January 2011 to May 2026 at daily frequency. The start date is chosen because iron-ore pricing shifted from annual benchmark contracts to spot-based pricing around 2010; restricting attention to the post-2010 era ensures the price series reflects a single, consistent pricing mechanism. After merging all series on common trading days the estimation sample contains 3,013 observations.

Equity prices for BHP, Rio Tinto and Fortescue are dividend- and split-adjusted closing prices from Yahoo Finance. We construct an equally weighted portfolio of the three firms' log prices as the primary dependent variable; this averages out firm-specific noise and isolates the sector-level commodity signal. The iron-ore price is the 62% Fe CFR China benchmark (USD per dry metric tonne), obtained as a clean weekday series. Copper futures serve as a secondary commodity in robustness tests. Control variables are the AUD/USD exchange rate, the VIX index of equity-market volatility, and the S&P 500 index as a measure of global equity-market conditions. The S&P 500 is preferred to a domestic index because it carries no mechanical overlap with the three sample firms, avoiding the partial endogeneity that an ASX-based control would introduce. All price series are expressed in natural logarithms.

Table 1 reports descriptive statistics for daily log returns. Iron-ore returns are far more volatile than the equity portfolio and display pronounced negative skewness and very heavy tails (excess kurtosis above 28), consistent with occasional large downward price moves. The mining portfolio itself shows mild negative skew and moderate excess kurtosis—the familiar fat-tailed profile of equity returns.

Table 1. Descriptive statistics, daily log returns (%), 2011–2026

Series	Mean	Std. dev.	Min	Max	Skew	Ex. kurt.
Portfolio	0.058	2.110	−10.82	10.47	−0.06	2.14
Iron ore	−0.015	1.993	−22.87	15.45	−1.93	28.39
Copper	0.011	1.708	−25.17	12.44	−1.13	19.70
AUD/USD	−0.012	0.739	−5.29	4.32	−0.31	3.17
VIX	−0.002	8.816	−44.24	101.91	1.49	12.19
S&P 500	0.059	1.209	−9.99	9.09	−0.80	9.31

N = 3,013. Returns are 100 × the first difference of log prices.

Figure 1. Iron-ore price (62% Fe CFR China, USD/t), 2011–2026, with major market events annotated.

Figure 1 plots the iron-ore price over the sample, annotated with the principal events of the period: the end of the supercycle bust and the early-2016 trough, the 2019 Vale dam-collapse supply shock, the COVID-19 crash, the 2021 price peak, the Evergrande-driven China property downturn, the Russia–Ukraine commodity shock, and more recent regulatory and trade events. The series is visibly non-stationary in levels with long swings, motivating the cointegration approach below.

3. Methodology

3.1 Unit-root testing

The NARDL bounds approach is valid when the variables are integrated of order zero or one, but not of order two. Augmented Dickey–Fuller tests (Table 2) confirm this. The log portfolio price, iron ore, copper and AUD/USD are non-stationary in levels but stationary in first differences (I(1)); the VIX and S&P 500 are stationary in levels (I(0)). No series is I(2), so the bounds framework applies.

Table 2. Augmented Dickey–Fuller unit-root tests

Variable	ADF (level)	p (level)	ADF (diff.)	p (diff.)	Order
ln Portfolio	−2.532	0.353	−15.003	< 0.01	I(1)
ln Iron ore	−2.542	0.349	−12.937	< 0.01	I(1)
ln Copper	−1.990	0.582	−15.300	< 0.01	I(1)
ln AUD/USD	−1.905	0.618	−15.083	< 0.01	I(1)
ln VIX	−5.294	< 0.01	−16.728	< 0.01	I(0)

Variable	ADF (level)	p (level)	ADF (diff.)	p (diff.)	Order
ln S&P 500	−3.684	0.025	−14.729	< 0.01	I(0)

Null hypothesis: a unit root. $p < 0.05$ rejects the null in favour of stationarity.

3.2 The NARDL model

The NARDL framework decomposes cumulative changes in the iron-ore log price into a partial sum of positive changes and a partial sum of negative changes:

$$x^{+}_{t} = \Sigma \max(\Delta x_i, 0), \quad x^{-}_{t} = \Sigma \min(\Delta x_i, 0).$$

These partial sums enter an error-correction model in which the lagged level terms capture the long-run relationship and the differenced terms capture short-run dynamics:

$$\Delta y_t = \alpha + \rho\, y_{t-1} + \vartheta^{+} x^{+}_{t-1} + \vartheta^{-} x^{-}_{t-1} + \gamma' z_{t-1} + \Sigma\, \varphi_j\, \Delta y_{t-j} + \Sigma\, \pi^{+}_{j}\, \Delta x^{+}_{t-j} + \Sigma\, \pi^{-}_{j}\, \Delta x^{-}_{t-j} + \varepsilon_t ,$$

where y is the log portfolio price, z the vector of controls (AUD/USD, VIX, S&P 500), and ρ the error-correction (speed-of-adjustment) coefficient. The implied long-run multipliers are $L^{+} = -\theta^{+}/\rho$ and $L^{-} = -\theta^{-}/\rho$. Because ρ is common to both, testing $\theta^{+} = \theta^{-}$ (long-run symmetry) is equivalent to testing $L^{+} = L^{-}$. We estimate the model by OLS and conduct all inference using Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey and West, 1987). This partial-sum specification follows established applications of the NARDL to asymmetric price transmission (Fousekis et al., 2016; Rezitis, 2019).

Lag orders are selected by the Bayesian Information Criterion over a grid of $p \in \{2,\ldots,5\}$ and $q \in \{0,1\}$; the BIC selects $p = 2$, $q = 1$. Crucially, this same selected specification is applied uniformly to every model in the paper—the main iron-ore model, the copper model, the market-control robustness models, and the firm-level models—so that comparisons across specifications are not confounded by differing lag structures.

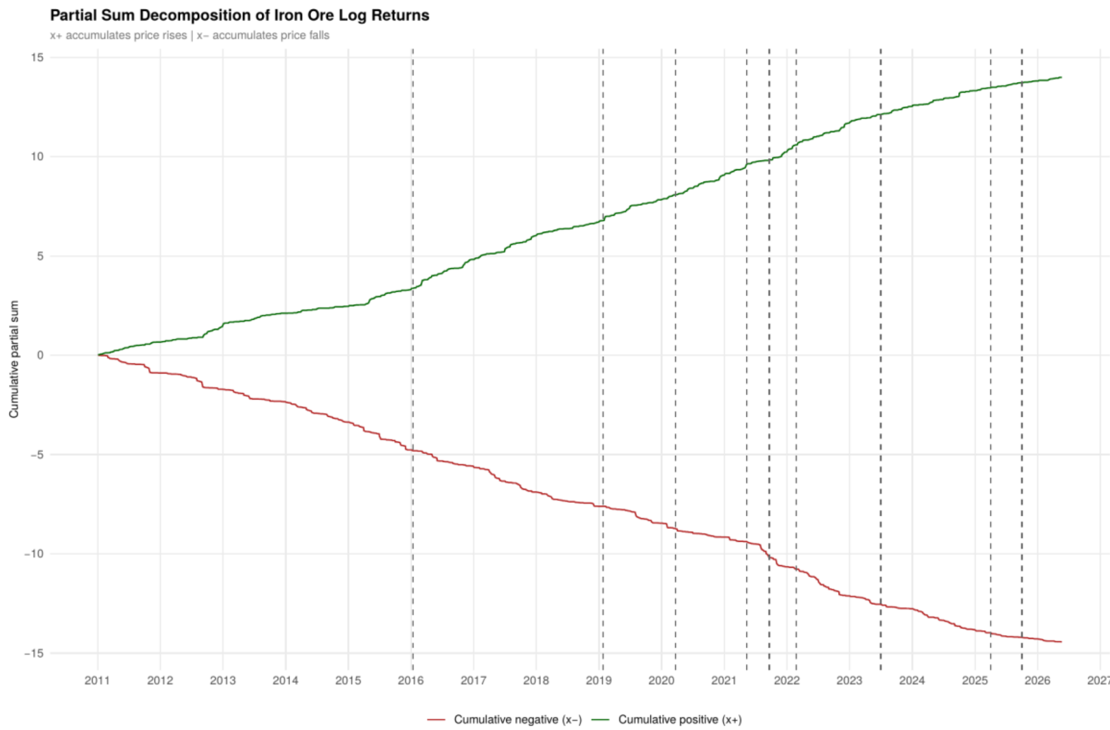

Figure 2. Partial-sum decomposition of cumulative iron-ore log returns into positive (x^+) and negative (x^-) components.

3.3 Bounds testing for a long-run relationship

We test for a long-run (cointegrating) relationship using the Pesaran, Shin and Smith (2001) bounds procedure, which provides two complementary statistics: a joint F-test on all lagged level terms and a t-test on the lagged dependent variable. Both are compared against critical-value bounds for the I(0) and I(1) cases. We report both, and discuss in Section 4.2 why their reading must be cautious in this application; in interpreting a significant error-correction term as evidence of a long-run relationship we follow Banerjee, Dolado and Mestre (1998).

3.4 Regime-switching extension

To capture discrete shifts between market states, we fit a two-state Markov-switching model to the portfolio's daily returns, in which the intercept, the autoregressive coefficient and the residual variance all switch between regimes (Hamilton, 1989). The model identifies a low-volatility ("calm") state and a high-volatility ("stress") state, together with the smoothed and filtered probabilities of being in each at every date. An ARCH-LM test on portfolio returns strongly rejects the null of no volatility clustering, motivating this regime structure; the presence of distinct volatility regimes in commodity and metals markets is well documented (Naeem et al., 2019).

We then embed the regime classification into the NARDL via a dummy-interaction specification on the full continuous sample. A stress dummy is constructed from the model's filtered probabilities—which condition only on information available up to each date, and so contain no look-ahead bias—and is interacted with both the long-run partial-sum terms and the short-run differenced terms. This preserves the full sample's lag structure exactly (unlike sample-splitting, which corrupts the lag relationships when regime days are non-consecutive), while letting both the long-run multipliers and the short-run transmission intensity differ across regimes. Because the stress dummy is itself an estimated object, we assess the robustness of the regime results to first-stage estimation uncertainty using a bootstrap described in Section 5.2.

4. Main results

4.1 Long-run asymmetry

Table 3 reports the main NARDL estimates. The error-correction coefficient is $\rho = -0.0160$ and is strongly significant (HAC $t = -4.17$, $p < 0.001$), confirming that the portfolio adjusts back toward its long-run relationship with iron-ore prices. The adjustment is slow: the implied half-life of a deviation is roughly $-\ln(0.5)/|\rho| \approx 43$ trading days, or about two months. Economically, this means that when the equities and the commodity drift apart the gap closes only gradually—a window of roughly two months over which a relative-value relationship between iron-ore futures and these equities can persist rather than arbitrage away within days. Both long-run partial-sum coefficients are positive and significant ($\theta^+ = 0.0115$, $p < 0.001$; $\theta^- = 0.0084$, $p = 0.005$).

Table 3. Main NARDL estimates (dependent variable: Δ ln portfolio)

Variable	Coef.	HAC SE	t	p
ln Portfolio$_{t-1}$ (ρ)	−0.0160	0.0038	−4.17	< 0.001
x^+ (iron ore)$_{t-1}$ (θ^+)	0.0115	0.0035	3.35	< 0.001
x^- (iron ore)$_{t-1}$ (θ^-)	0.0084	0.0030	2.80	0.005
Δx^+ (iron ore) (π^+)	0.342	0.052	6.56	< 0.001
Δx^- (iron ore) (π^-)	0.206	0.035	5.94	< 0.001
ln AUD/USD$_{t-1}$	0.0046	0.0074	0.62	0.533
ln VIX$_{t-1}$	0.0007	0.0014	0.47	0.641
ln S&P 500$_{t-1}$	−0.0047	0.0043	−1.09	0.278

Newey–West HAC standard errors. Short-run own-lag and lagged-difference terms included but omitted here for brevity. N = 3,013.

The implied long-run multipliers are $L^+ = 0.720$ and $L^- = 0.523$. A Wald test of the symmetry restriction $\theta^+ = \theta^-$ is decisively rejected ($F = 13.90$, $p = 0.0002$): a sustained iron-ore price increase moves the mining portfolio's long-run equilibrium value substantially more than an equally sized decrease. Figure 3 displays the two multipliers and the 0.197 gap between them.

The economic content of this gap is a convex, option-like response of mining-equity value to the commodity. A 10% sustained rise in the iron-ore price is associated with roughly a 7.2% rise in the portfolio's long-run equilibrium value, whereas a 10% fall is associated with only about a 5.2% decline—the equities capture more of the upside than the downside. This is the pattern operating leverage and the equity-as-call-option-on-assets view would predict: with largely fixed extraction costs, a price rise flows disproportionately to the residual claim, while on the downside the option value of the firm's assets and the scope to defer or curtail output cushion equity holders. For an investor, the immediate consequence is that exposure to iron ore through these equities is not linear—a hedge constructed on the assumption of symmetric pass-through will be systematically mis-sized, over-hedging the downside and under-hedging the upside.

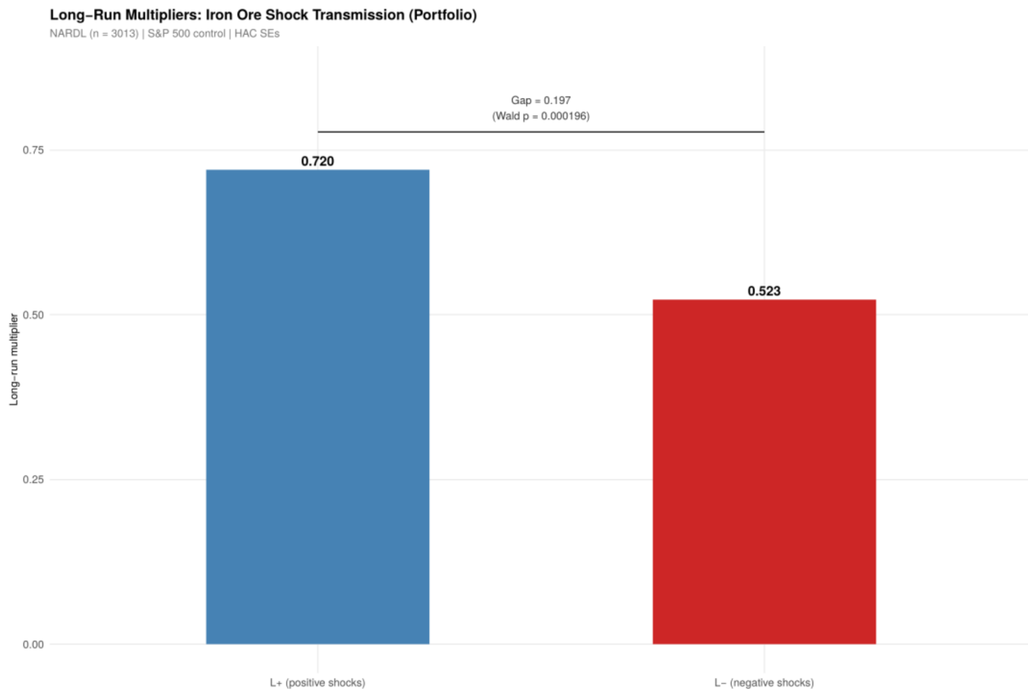

Figure 3. Long-run multipliers for positive (L^+) and negative (L^-) iron-ore shocks, with the asymmetry gap and Wald-test p-value.

4.2 The long-run relationship: a cautious reading

The strength of the cointegration evidence requires care. The bounds F-statistic is 3.66. Under the Pesaran–Shin–Smith Case III critical values for five long-run forcing variables, the 5% I(1) upper bound is 3.79; our statistic therefore falls between the I(0) and I(1) bounds and is inconclusive at the 5% level, though it clears the 10% bound (3.35). The bounds t-statistic on ρ is −4.14, which likewise sits between the 5% bounds but clears the 10% I(1) threshold (−3.86).

We therefore describe the long-run evidence as suggestive rather than conclusive. Two considerations support a long-run interpretation despite the inconclusive F. First, the error-correction coefficient is negative and strongly significant ($t = -4.17$), which in the Banerjee–Dolado–Mestre tradition is itself evidence of error correction toward a long-run relationship. Second, the bounds t-test clears its 10% threshold. The slowness of adjustment (a two-month half-life) and the marginal F are two sides of the same coin—a small $|\rho|$ means both a slowly converging equilibrium and a relationship that is only modestly identified by the joint F-test. We prefer to state this transparently rather than over-claim cointegration on a single marginal statistic.

4.3 Robustness to the market control

Because the choice of broad-market control is a legitimate concern—a domestic index would mechanically contain the sample firms—we re-estimate the model under three controls: none (AUD/USD and VIX only), the S&P 500 (our main specification, with no overlap), and the ASX 200 (included only for comparison, despite its endogeneity). Table 4 shows the asymmetry is robust throughout: L^+ exceeds L^- in every specification, with gaps of 0.16–0.22 and Wald p-values at or below 0.0002. Figure 4 displays the multipliers across the three controls.

Table 4. Long-run multipliers across market-control specifications

Specification	L^+	L^-	Gap	Wald p	Bounds F
No market control	0.710	0.547	0.163	0.0001	4.17
S&P 500 (main)	0.720	0.523	0.197	0.0002	3.66
ASX 200 (comparison)	0.767	0.549	0.218	< 0.0001	4.70

Bounds F to be read against the 5% I(1) value of 3.79. The main (S&P 500) specification is the most conservative on cointegration because the I(0) control absorbs common variation, lowering the joint F.

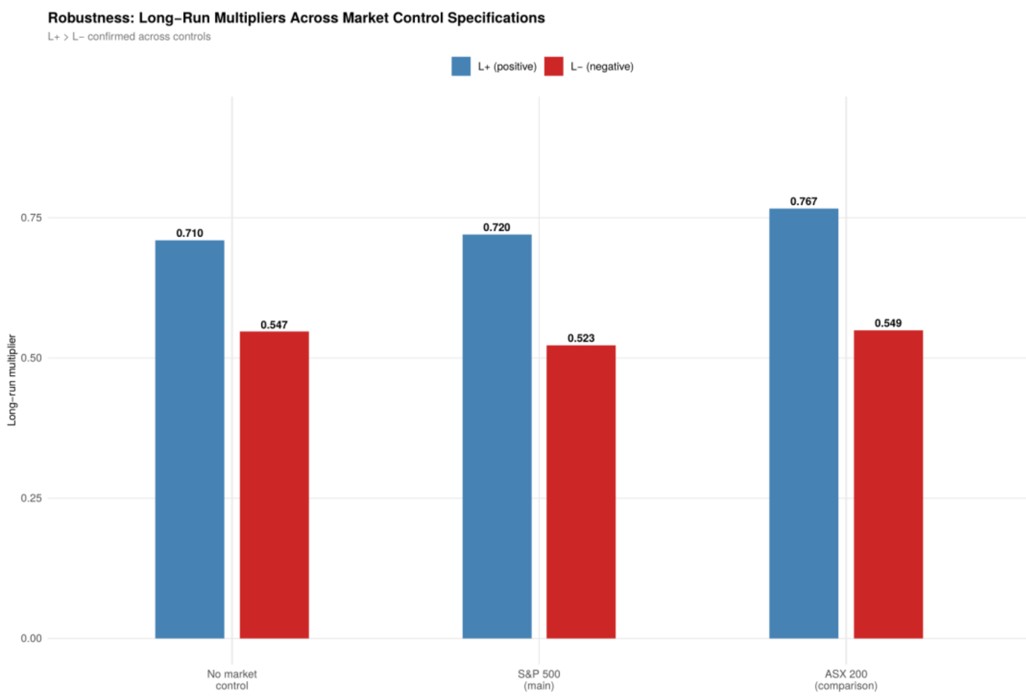

Figure 4. Long-run multipliers under alternative market-control specifications. $L^+ > L^-$ in all three.

4.4 Firm-level heterogeneity

If iron-ore exposure is the operative channel, the asymmetry should be strongest in the most iron-ore-concentrated firm. We estimate the model separately for each producer (Table 5). All three exhibit significant asymmetry with $L^+ > L^-$. The pattern by business model is striking: Fortescue, an effectively pure-play iron-ore producer, shows long-run multipliers three to four times those of the diversified majors—$L^+ = 1.314$ against 0.389 (BHP) and 0.334 (Rio Tinto)—and roughly double their asymmetry gap.

Table 5. Firm-level NARDL estimates

Firm	L^+	L^-	Gap	Wald p
BHP (diversified)	0.389	0.216	0.173	0.003
Rio Tinto (diversified)	0.334	0.198	0.136	0.008

Firm	L^+	L^-	Gap	Wald p
Fortescue (pure-play)	1.314	0.968	0.346	0.005
Portfolio (equal weight)	0.720	0.523	0.197	< 0.001

We are careful not to over-interpret the ordering among the diversified pair. BHP's multiplier slightly exceeds Rio Tinto's despite broadly comparable iron-ore shares, so the data do not support a strictly monotonic gradient across all three firms. What they support is a clear pure-play-versus-diversified contrast: the firm whose cash flows are almost entirely iron-ore-driven responds three to four times as strongly as its diversified peers, while the two diversified majors are statistically close to one another. This is exactly the pattern expected if iron-ore exposure drives the asymmetric transmission, and echoes the sector-dependence of commodity–equity responses documented by Arouri and Nguyen (2010) and the exposure-concentration finding of Mishra and Acharya (2025). Figure 5 shows the firm-level multipliers alongside the portfolio.

This heterogeneity is directly useful for portfolio construction. The three firms are not interchangeable proxies for "Australian mining": Fortescue is effectively a leveraged, asymmetric bet on the iron-ore price, transmitting roughly three to four times the commodity move of BHP or Rio Tinto, whereas the diversified majors dilute that exposure through their other commodity and operating segments. An investor seeking clean, high-beta exposure to iron ore finds it in the pure-play producer; an investor seeking mining-sector participation while limiting concentrated single-commodity risk is structurally better served by the majors. The same logic applies in reverse to risk management: a desk hedging Fortescue exposure must size its iron-ore hedge several times larger, per dollar of equity, than for the majors—and, given the asymmetry documented above, must size the long and short legs of that hedge differently.

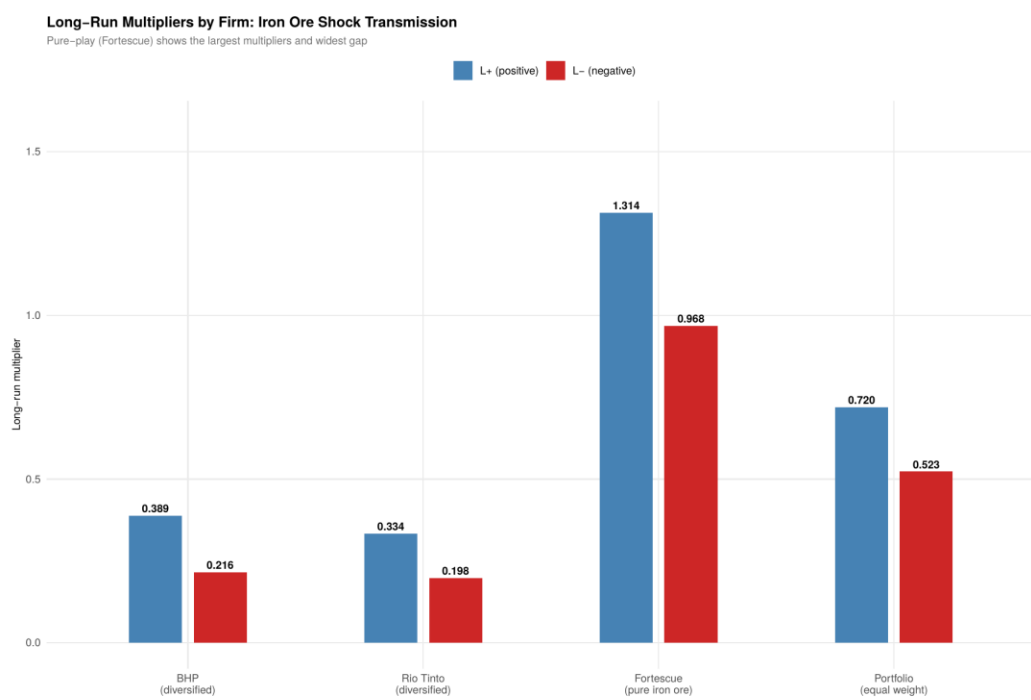

Figure 5. Long-run multipliers by firm. The pure-play producer (Fortescue) shows the largest multipliers and the widest L^+–L^- gap.

4.5 Copper as a secondary commodity

Re-estimating the model with copper in place of iron ore provides a check on whether the asymmetry is specific to iron ore or a more general commodity–equity feature. The copper specification yields $L^+ = 0.950$ and $L^- = 0.628$ with a significant symmetry rejection (Wald $F = 5.64$, $p = 0.018$), corroborating the direction of the asymmetry. However, the individual long-run copper coefficients are not statistically significant, and copper's error-correction coefficient (−0.0043) is much smaller than iron ore's. Copper therefore reinforces the directional finding but offers weaker evidence of a robust long-run copper–equity relationship—consistent with iron ore being the dominant fundamental for these firms.

5. Regime-conditional transmission

5.1 Two regimes, and a two-part result

The Markov-switching model identifies two clearly distinct states. The stress regime has a daily return standard deviation of 3.00% and a near-zero mean, and accounts for 31.1% of trading days; the calm regime has a standard deviation of 1.55%, a small positive mean (+0.084% per day), and accounts for the remaining 68.9%. The regimes are moderately persistent, with expected durations of roughly 17 and 34 trading days respectively. Figure 6 plots the smoothed probability of the stress regime, which rises sharply around each of the sample's major risk events.

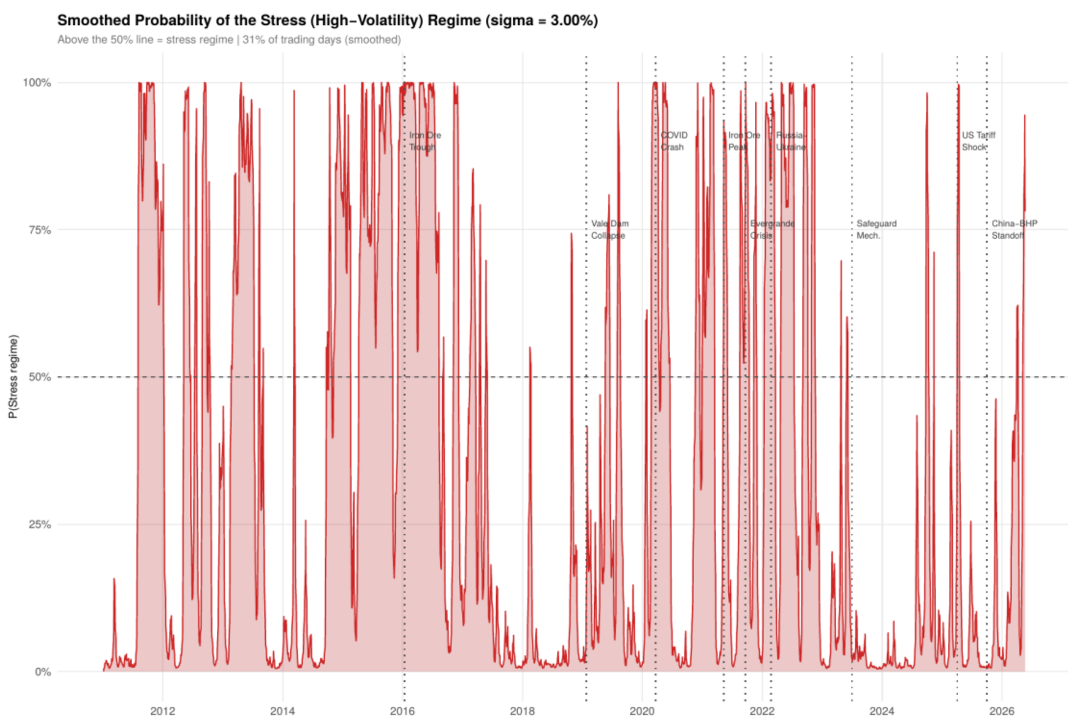

Figure 6. Smoothed probability of the high-volatility (stress) regime, 2011–2026.

Embedding the regime classification in the NARDL produces a clean two-part result, displayed together in Figure 7. In the long run (left panel), the asymmetry is essentially identical across regimes: the calm-regime gap is 0.191 ($L^+ = 0.774$, $L^- = 0.582$; Wald $F = 17.76$, $p < 0.001$) and the stress-regime gap is

0.198 ($L^+ = 0.635$, $L^- = 0.437$; Wald $F = 16.75$, $p < 0.001$). A direct test of whether the asymmetry differs across regimes cannot reject equality ($F = 0.24$, $p = 0.624$). The long-run asymmetry is, in this sense, regime-invariant—a structural feature of the relationship rather than a crisis phenomenon. Both multipliers do fall somewhat in the stress regime, consistent with global risk factors crowding out commodity-specific pricing when markets are turbulent—a state-dependence of commodity–equity spillovers also documented, in a connectedness framework, by Mensi et al. (2021)—but the proportional gap between them is unchanged.

The short run (right panel) tells a different story. The same-day transmission of iron-ore shocks intensifies sharply in the stress regime: the positive-shock coefficient rises from 0.184 in calm periods to 0.536 in stress, and the negative-shock coefficient from 0.113 to 0.365—approximately a tripling in both directions. The two regime-interaction terms are individually significant (positive: 0.353, $p = 0.0015$; negative: 0.253, $p = 0.0014$). Importantly, the increase is of similar proportion in both directions: the within-regime short-run asymmetry is not statistically significant in either state, and the short-run regime-difference in asymmetry is insignificant ($p = 0.49$). The correct characterisation is therefore that short-run transmission intensity rises in stress for both directions roughly equally—the immediate magnitude of transmission is state-dependent, while the asymmetry itself is not.

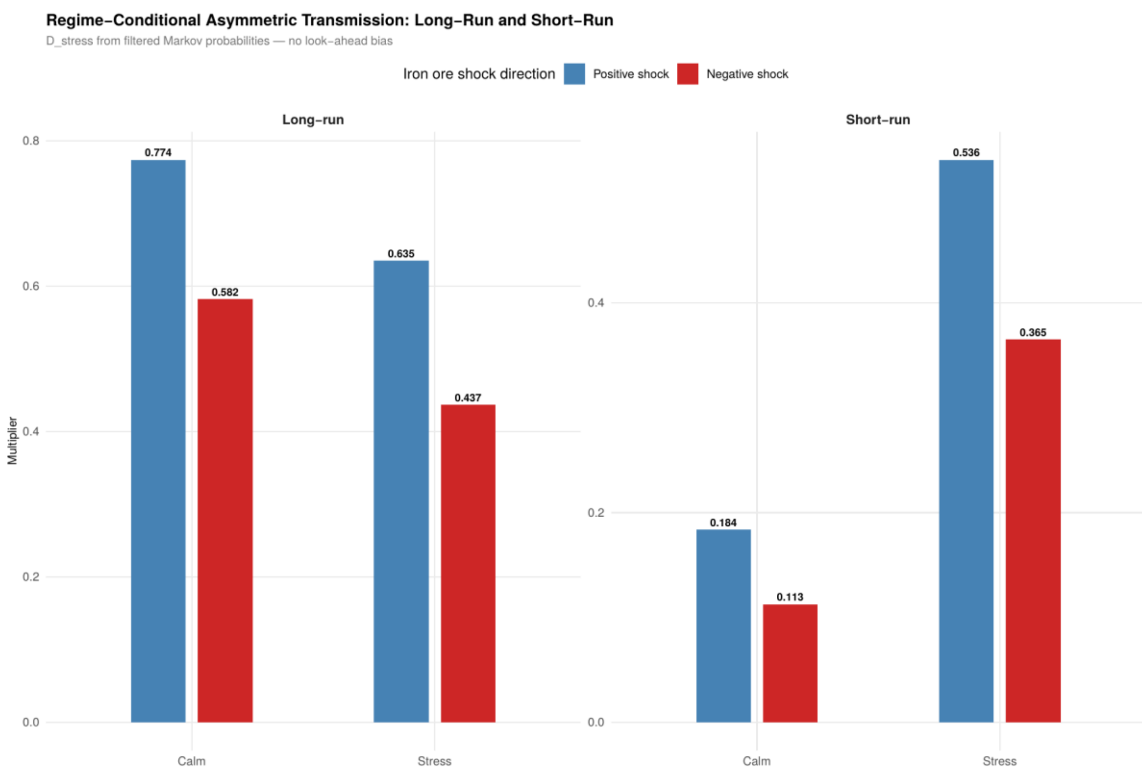

Figure 7. Regime-conditional transmission. Left: long-run multipliers, with a near-identical asymmetry gap in calm and stress. Right: short-run coefficients, which roughly triple from calm to stress in both directions. Regimes from filtered Markov probabilities (no look-ahead bias).

5.2 Robustness to first-stage estimation uncertainty

Because the stress dummy is generated from an estimated Markov model, conventional standard errors in the second-stage NARDL understate uncertainty: they treat the regime classification as known with certainty. We address this with a two-stage moving-block bootstrap that re-estimates the entire pipeline—resampling the data in blocks, refitting the Markov model, reconstructing the stress dummy, and re-estimating the interaction NARDL on each replication—so that first-stage uncertainty propagates into the second-stage inference. Table 6 compares the resulting standard errors with a simpler classification-only bootstrap.

Table 6. Bootstrap standard errors for the regime-interaction terms

Quantity	Estimate	SE (HAC)	SE (full BS)	95% CI (full BS)
δ^+ (long-run, positive)	−0.0026	0.0027	0.0039	[−0.0086, 0.0059]
δ^- (long-run, negative)	−0.0027	0.0027	0.0038	[−0.0088, 0.0062]
Regime difference ($\delta^+ - \delta^-$)	0.0001	0.0001	0.0005	[−0.0012, 0.0010]

Full BS = two-stage moving-block bootstrap (block length 21; 196 usable replications) that propagates Markov first-stage uncertainty.

Two points emerge. First, propagating first-stage uncertainty roughly triples the standard errors on the regime-interaction terms relative to the naive HAC values—vindicating the concern that treating the regime dummy as known would overstate precision. Second, and decisively for our interpretation, the regime-difference in the asymmetry remains an unambiguous and well-identified zero: its bootstrap confidence interval, [−0.0012, 0.0010], is tightly centred on zero even under this most demanding treatment. The regime-invariance of the long-run asymmetry is thus not an artefact of understated standard errors—it survives a bootstrap explicitly designed to inflate them.

6. Economic significance and implications

The statistical results above establish that the asymmetry is real, robust and structural. This section draws out what it means for the firms, their investors and the broader market, and why the findings matter beyond the econometrics.

6.1 Interpreting the asymmetry: candidate mechanisms

The central empirical message is that mining-equity value responds convexly to the iron-ore price: good commodity news is transmitted more fully than bad news. The NARDL establishes that this asymmetry exists and is robust, but it does not by itself identify what produces it. We therefore offer the following as candidate explanations consistent with the evidence, rather than as mechanisms the model tests directly. Two are most natural given the structure of these businesses. The first is operating leverage: iron-ore producers carry largely fixed extraction and infrastructure costs, so a price increase would be expected to pass through to operating margins and free cash flow more than proportionately, while a price fall is partly absorbed before it reaches the bottom line. The second is the option-like character of the equity claim: shareholders hold what is effectively a call option on the firm's assets, and the ability to defer development, curtail high-cost output, or draw down inventory would cushion equity value on the downside in a way that has no symmetric counterpart on the upside. Other channels could contribute—asymmetric expectation formation, or greater investor attention to good than to bad commodity news—

and we cannot adjudicate among them here. What we can say is that the firm-level pattern is consistent with the exposure-based reading: the most commodity-concentrated firm (Fortescue) shows the largest asymmetry, which is what both the operating-leverage and option-value stories would predict if iron-ore exposure is the operative channel. We treat this as suggestive support for the interpretation, not as confirmation of a specific mechanism; isolating the channel is left to the future work noted in Section 8.

6.2 Implications for hedging and risk management

The most direct practical consequence is for anyone hedging mining-equity exposure with iron-ore derivatives, or vice versa. A hedge built on the implicit assumption of symmetric transmission—the default in most linear factor models—will be miscalibrated in a predictable direction: it will tend to over-hedge against price declines and under-hedge against price rises, because the true equity sensitivity is larger on the upside ($L^+ = 0.72$) than the downside ($L^- = 0.52$). A risk manager who recognises the asymmetry can set the two legs of the hedge differently and reduce this basis risk. The firm-level results sharpen the point: because Fortescue transmits roughly three to four times the commodity move of the majors, hedge ratios cannot be transplanted across the three stocks, and a portfolio that treats them as a single "iron-ore" block will carry substantially more concentrated commodity risk than its nominal sector weighting suggests.

6.3 Implications for portfolio construction and the regime split

For asset allocation, the heterogeneity across firms is itself the actionable result. The three producers span a spectrum of commodity beta: Fortescue offers concentrated, high-sensitivity exposure for an investor who wants iron-ore upside, while BHP and Rio Tinto offer diluted exposure more suitable for an investor seeking sector participation without single-commodity concentration. The regime results then separate two questions that are easy to conflate. Because the long-run asymmetry is regime-invariant, the hedge ratios and exposure estimates an investor builds in calm periods remain valid through stress periods—the structural relationship does not break down when it is most needed. But because short-run transmission intensity roughly triples in the stress regime, the speed and size of the immediate impact rise sharply when markets are turbulent. The implication is operational: the level of a hedge can be set on stable long-run parameters, but its monitoring and rebalancing frequency should increase in stressed states, when commodity shocks hit equity value faster and harder even though the underlying asymmetry is unchanged.

6.4 Broader relevance

Finally, the findings speak to the resource-dependence of the Australian market as a whole. The iron-ore majors are among the largest constituents of the domestic index, so an asymmetry in how they price commodity news propagates into broad-market and superannuation portfolios that most Australian investors hold by default. Understanding that this exposure is convex and state-dependent—rather than the linear, constant relationship a standard beta would imply—is relevant not only to specialist commodity investors but to anyone with passive exposure to the market. More generally, the two-part structure we document—a stable asymmetric relationship whose immediate intensity varies with the market state—offers a template for thinking about other commodity–equity pairs and, prospectively, about how carbon-price and transition risks may come to be transmitted into resource-sector valuations.

7. Diagnostics

The model's residuals show serial correlation (Breusch–Godfrey LM(4) = 16.46, $p = 0.002$), which is why all inference uses HAC standard errors. A Ramsey RESET test rejects the null of correct functional form ($F = 16.62$, $p < 0.001$). We take this seriously but interpret it in light of three considerations rather than treating it as decisive.

First, the RESET test is constructed from powers of the fitted values, which in a NARDL error-correction model blend I(1) levels with I(0) differences and already embed the asymmetric partial-sum structure by design. Its powers can therefore re-detect the model's intended nonlinearity and flag it as misspecification, and the test cannot distinguish modelled nonlinearity from omitted nonlinearity in this setting—a known limitation of applying standard diagnostics to NARDL models. This concern is not merely conjectural: Monte Carlo evidence shows that the power of the RESET test is governed by the degree of nonlinearity in the underlying relationship (Christodoulou-Volos and Tserkezos, 2023), so a strongly nonlinear specification such as ours is precisely the case in which a rejection is most likely to reflect the model's own structure. A rejection also identifies only that some functional-form issue may be present, not its source or remedy, which limits how much weight it can bear here. Second, the reported results are empirically invariant to the short-run dynamics the test might be reacting to: across all eight (p, q) lag specifications in the selection grid, L^+ lies in 0.708–0.740, L^- in 0.506–0.544, the gap in 0.196–0.203, and the symmetry Wald test rejects with $p \le 0.0005$ throughout (Table 7). If RESET were detecting genuine under-fitting that contaminated the estimates, enriching the dynamics would move the multipliers; it does not. Third, the model's parameters are stable over the full sample: an OLS-CUSUM test does not reject parameter constancy ($S_0 = 0.487$, $p = 0.972$), as Figure 8 shows.

Table 7. Lag-robustness grid: long-run estimates across (p, q) specifications

p	q	L^+	L^-	Gap	Wald p	RESET p
2	0	0.740	0.544	0.196	0.0001	< 0.001
3	0	0.740	0.543	0.197	0.0001	< 0.001
4	0	0.735	0.534	0.201	0.0002	< 0.001
5	0	0.731	0.532	0.199	0.0002	< 0.001
2	1	0.720	0.523	0.197	0.0002	< 0.001
3	1	0.719	0.520	0.198	0.0003	< 0.001
4	1	0.711	0.508	0.203	0.0005	< 0.001
5	1	0.708	0.506	0.202	0.0004	< 0.001

BIC selects (p = 2, q = 1). Multipliers and the symmetry test are stable across the grid while RESET rejects throughout—indicating the rejection does not contaminate the reported estimates.

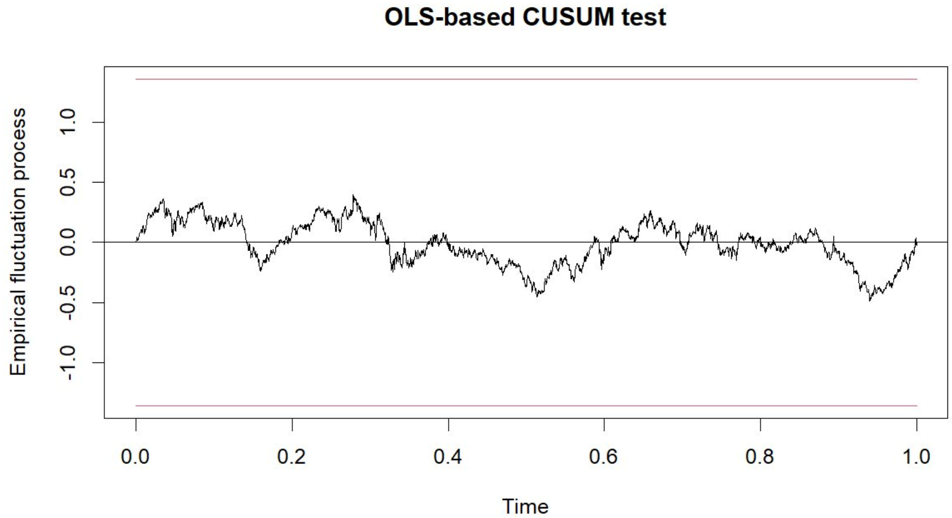

Figure 8. OLS-CUSUM test of parameter stability. The cumulative-sum process remains within the 5% bands throughout the sample (S_0 = 0.487, p = 0.972).

8. Limitations and future research

Several limitations qualify the results and point to natural extensions. We state them plainly, as each bears on how the findings should be used.

First, the evidence for a long-run cointegrating relationship is suggestive rather than conclusive. As discussed in Section 4.2, the bounds F-statistic is inconclusive at the 5% level under the correct critical values; the long-run interpretation rests on the significantly negative error-correction coefficient and the bounds t-test clearing its 10% threshold. Readers should treat the long-run multipliers as well-estimated conditional on a long-run relationship existing, while recognising that the existence of that relationship is supported but not established beyond doubt.

Second, the Ramsey RESET test rejects the null of correct functional form. We argue in Section 7 that this largely reflects the test re-detecting the model's own designed nonlinearity rather than genuine misspecification, and we show the multipliers are stable across the full lag grid while RESET continues to reject. The argument is, in our view, persuasive, but it is an argument rather than a clean diagnostic pass, and we present it as such.

Third, the regime analysis is two-stage: the stress dummy is generated from an estimated Markov-switching model and then entered into the NARDL, rather than the two components being estimated jointly. We address the resulting generated-regressor problem with a two-stage block bootstrap that propagates first-stage uncertainty, and the regime-invariance conclusion survives it. Nonetheless, a fully joint estimation of the regime process and the asymmetric transmission would be a more efficient, if considerably more complex, approach, and we leave it to future work.

Fourth, the NARDL identifies the asymmetry but does not by itself isolate the mechanism behind it. The economic interpretation in Section 6—operating leverage and the option-like character of the equity claim—is consistent with the evidence and with the firm-level pattern, but the model does not directly test these channels against alternatives such as asymmetric expectation formation or differential investor attention to good and bad commodity news. Distinguishing among the candidate mechanisms, perhaps using firm-level cost structures or options-market data, is a promising direction.

Fifth, the scope of the study is deliberately narrow: three Australian producers, a single commodity, and the post-2010 spot-pricing era. This focus is what allows a clean identification, but it limits external validity. Whether the same asymmetric, regime-invariant structure holds for other commodity–equity pairs, other markets, or earlier pricing regimes is an empirical question our design cannot answer. The main analysis also uses an equally weighted portfolio to isolate the sector-level commodity signal; we address firm-level heterogeneity directly in Section 4.4, though a fuller cross-sectional or panel treatment remains open. Finally, the daily frequency captures both short-run dynamics and slow long-run adjustment but is silent on intraday transmission and on structural change at horizons longer than the sample.

None of these qualifications overturns the central findings, which are robust across the checks we are able to run. They do, however, mark the boundary of what the present design supports, and they map the most useful directions for extending it.

9. Conclusion

This paper documents a robust long-run asymmetry in the transmission of iron-ore price shocks to Australian mining equities. Positive iron-ore movements raise the sector's long-run equilibrium value by more than equally sized negative movements lower it—long-run multipliers of 0.720 versus 0.523, a gap that is highly significant and robust across market controls and lag specifications. The asymmetry is concentrated by business model: the pure-play producer responds three to four times as strongly as the diversified majors, precisely as an iron-ore-exposure channel would predict.

The regime analysis sharpens the interpretation into a two-part structure. The long-run asymmetry is regime-invariant: statistically indistinguishable between calm and stress states, and robust to a two-stage block bootstrap that propagates the uncertainty of the estimated regime classification. The short-run intensity of transmission, by contrast, roughly triples in the stress regime in both directions. The asymmetric structure of commodity–equity transmission is thus a stable feature of the relationship, while its immediate magnitude is state-dependent.

We have been deliberately cautious about the cointegration evidence, resting the long-run interpretation on a significantly negative error-correction term and the bounds t-test rather than on a marginal F-statistic, and deliberately careful about the firm-level pattern, claiming a pure-play-versus-diversified contrast rather than a strict monotonic gradient. These choices reflect what the data support. As set out in Section 6, the findings carry concrete implications: the convex, asymmetric response calls for asymmetric hedge sizing; the firm-level heterogeneity makes the three producers distinct instruments for commodity exposure rather than interchangeable proxies; and the two-part regime structure means hedge ratios can be set on stable long-run parameters while rebalancing should intensify in stressed states. A natural extension is to apply the same framework to other commodity–equity pairs and to carbon-market linkages, where asymmetric and state-dependent transmission may be similarly important.

References

[1] Arouri, M. E. H., & Nguyen, D. K. (2010). Oil prices, stock markets and portfolio investment: Evidence from sector analysis in Europe over the last decade. Energy Policy, 38(8), 4528–4539.

[2] Banerjee, A., Dolado, J., & Mestre, R. (1998). Error-correction mechanism tests for cointegration in a single-equation framework. Journal of Time Series Analysis, 19(3), 267–283.

[3] Christodoulou-Volos, C., & Tserkezos, D. (2023). Sensitivity of the Ramsey's regression specification error term test on the degree of nonlinearity of the functional form. Journal of Applied Economic Sciences, 18(1), 5–10.

[4] Fousekis, P., Katrakilidis, C., & Trachanas, E. (2016). Vertical price transmission in the US beef sector: Evidence from the nonlinear ARDL model. Economic Modelling, 52, 499–506.

[5] Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica, 57(2), 357–384.

[6] Mensi, W., Al Rababa'a, A. R., Vo, X. V., & Kang, S. H. (2021). Asymmetric spillover and network connectedness between crude oil, gold, and Chinese sector stock markets. Energy Economics, 98, 105262.

[7] Mishra, L., & Acharya, R. H. (2025). The asymmetric interaction between oil price change and stock returns of the renewable energy companies in India: A panel NARDL approach. Asia-Pacific Financial Markets, 32(1), 103–116.

[8] Naeem, M., Tiwari, A. K., Mubashra, S., & Shahbaz, M. (2019). Modeling volatility of precious metals markets by using regime-switching GARCH models. Resources Policy, 64, 101497.

[9] Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica, 55(3), 703–708.

[10] Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. Journal of Applied Econometrics, 16(3), 289–326.

[11] Rezitis, A. N. (2019). Investigating price transmission in the Finnish dairy sector: An asymmetric NARDL approach. Empirical Economics, 57(3), 861–900.

[12] Shin, Y., Yu, B., & Greenwood-Nimmo, M. (2014). Modelling asymmetric cointegration and dynamic multipliers in a nonlinear ARDL framework. In R. Sickles & W. Horrace (Eds.), Festschrift in Honor of Peter Schmidt (pp. 281–314). Springer.

[13] Truong, L. D., Nguyen, A. T. K., Friday, H. S., & Doan, N. T. (2024a). The asymmetric effects of oil prices on stock returns: Evidence from Hanoi Stock Exchange, Vietnam. International Journal of Energy Economics and Policy, 14(5), 236–241.

[14] Truong, L. D., Nguyen, A. T. K., Friday, H. S., & Doan, N. T. (2024b). The asymmetric effects of oil price volatility on stock returns: Evidence from Ho Chi Minh Stock Exchange. Journal of Risk and Financial Management, 17(7), 261.